

Lab 3 – Architectural Justification: Household Carbon Footprint & Target Tracker

Architecture Selection: We chose Layered Architecture for the Household Carbon Footprint & Target Tracker System.

Architectural Choice

Layered Architecture (Presentation → Business Logic → Data) is the selected style. The system is decomposed into five components: User Interface Component (Presentation Layer), Consumption Logger Component and Emission Calculator Component (Business Layer), Target Tracker Component (Business Layer), and Database Component (Data Layer).

Reason 1 – Sequential Data Flow Matches the System's Core Logic

The system's core workflow is strictly sequential: a resident logs monthly utility data, the system converts it to CO₂e using EPA factors, and the result updates historical records and reduction targets. This linear dependency chain maps directly onto a layered architecture. The Presentation Layer captures input, the Business Layer runs the emission calculation and target evaluation, and the Data Layer persists results. Microservices would add unnecessary inter-service coordination overhead for what is fundamentally a straight-through processing pipeline for a single-household tool.

Reason 2 – External Conversion Factor Integration is Well-Contained

The system must integrate with an external EPA Conversion Factor source to produce accurate CO₂e values. In a layered model, the Emission Calculator Component in the Business Layer owns this dependency through a well-defined Emission Calculation Interface. This isolates the external coupling to a single component — if EPA factor formats change, only the calculator is affected. Client-Server architecture would not cleanly separate this concern, and microservices would unnecessarily expose it as an independent service endpoint for a system that has no need for independent scaling.

Security Advantage

Layered architecture enforces strict access control between tiers. The Presentation Layer has no direct path to the Database Component — all reads and writes flow through the Business Layer components. This means household energy consumption data and personal reduction targets are never directly accessible from the UI, reducing exposure of sensitive household activity data to potential front-end vulnerabilities or unauthorized direct database access.

Performance Benefit

The Emission Calculator Component can cache EPA conversion factors in the Business Layer after the first fetch, so subsequent monthly calculations do not require repeated external lookups. Additionally, because all layers run in-process, the historical emissions chart specified in NFR-001 renders from local database queries without any network round-trip latency — directly supporting the sub-200ms rendering requirement identified in Lab 1.